

Skills Summary

Common Denominators for Unlike Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Rewriting a Fraction with a New Denominator

Rule — multiply top and bottom by the same number. Whatever you multiply the denominator by, multiply the numerator by too. The value of the fraction does not change, only the size of the pieces. To find the factor, ask: *what times the old denominator gives the new one?*

Example: Rewrite $\frac{2}{5}$ with a denominator of 20.

Step 1. The denominator has to become 20, so first find what the old denominator was multiplied by.

Step 2. $20 \div 5 = 4$, so multiply the numerator by 4 as well: 2×4 .

$\Rightarrow$ new numerator = 8

Try it: Rewrite $\frac{3}{7}$ with a denominator of 21. What is the new numerator?

check: $\frac{6}{21}$

(!) Watch out: never *add* to the denominator. Going from sevenths to twenty-firsts is multiplying by 3, not adding 14, and the numerator must be multiplied by the very same 3.

Skill 2 — Adding and Subtracting Unlike Fractions

Rule. You can only add or subtract fractions whose pieces are the same size. Find a common denominator (any common multiple works; the *least* one keeps the numbers small), rewrite both fractions with it, then add or subtract the numerators only — the denominator stays put. Simplify at the end.

Example: Add $\frac{1}{3} + \frac{1}{4}$

Step 1. Thirds and quarters are different-sized pieces, so nothing can be added until both are written over one common denominator.

Step 2. 3 and 4 share no factor, so use 12: $\frac{1}{3} = \frac{4}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Now add the numerators.

$\Rightarrow \frac{7}{12}$

Try it: Subtract $\frac{3}{4} - \frac{2}{5}$

check: $\frac{20}{20}$

Skill 3 — Mixed Numbers with Unlike Denominators

Rule. Handle the whole numbers and the fraction parts separately. Give the fraction parts a common denominator first. When subtracting, if the top fraction is the smaller one, *regroup*: borrow one whole and add it to the fraction part as a full set of those pieces.

Example: Add $1\frac{1}{4} + 2\frac{2}{3}$

Step 1. Whole numbers add straight away; the fraction parts are quarters and thirds, so they need a common denominator before they can be combined.

Step 2. $1 + 2 = 3$, and with denominator 12: $\frac{1}{4} = \frac{3}{12}$, $\frac{2}{3} = \frac{8}{12}$, so the fraction part is $\frac{11}{12}$.

$$\Rightarrow 3\frac{11}{12}$$

Try it: Subtract $3\frac{1}{2} - 1\frac{1}{3}$

$\frac{9}{1}$ check